

Performance Comparison of OFDM with 256-QAM and QPSK Modulation Schemes Using Simulink

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ABSTRACT

A Matlab Simulation program has been written to investigate Orthogonal Frequency Division Multiplexing (OFDM) incorporated in communication channels using 256-QAM and QPSK for encoding. Single carrier QAM and PSK and multicarrier OFDM are compared to demonstrate the strength of OFDM in multipath channels. Performance metrics such as Bit Error Rate, Spectral efficiency and Signal to Noise Ratio will be analysed and compared between the different modulation techniques. OFDM in the long run provides a significant change in spectral efficiency and robustness to multipath fading. These findings become useful in choosing modulation schemes that are suitable to our communication channel.

Index Terms: A.0.1 [QAM, PSK, OFDM, Modulation, Spectral efficiency, Bit Error Rate, Signal to Noise Ratio]:

1 INTRODUCTION

Orthogonal Frequency Division Multiplexing (OFDM) has become a cornerstone of modern communication systems, especially in environments characterized by multipath propagation [1]. Its resilience to frequency-selective fading and its ability to achieve high data rates have made it a critical technology for applications ranging from wireless local area networks (WLANs) to digital video broadcasting (DVB). This report presents a detailed investigation into the performance of OFDM when employed in communication channels using 256-QAM and M-PSK modulation schemes. A dedicated MATLAB simulation program has been developed to facilitate this analysis, enabling a direct comparison between OFDM and traditional single-carrier modulation techniques, namely QAM and PSK. This comparison is crucial for understanding the performance trade-offs associated with each modulation method in the presence of multipath. The investigation focuses on three key performance indicators: Bit Error Rate (BER), which quantifies the reliability of the communication link; Spectral Efficiency, which measures how effectively the available bandwidth is utilized; and Signal-to-Noise Ratio (SNR), which represents the power of the desired signal relative to the background noise [2]. By meticulously analyzing these metrics across the different modulation schemes, this study aims to demonstrate the specific strengths of OFDM in mitigating the detrimental effects of multipath fading, improving overall system performance, and achieving higher spectral efficiency. The insights gained from this comparative analysis are intended to provide valuable guidance for selecting the most appropriate modulation technique for a given communication channel, considering its specific characteristics and performance requirements. Ultimately, this work contributes to a deeper understanding of OFDM and its role in optimizing communication system design.

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2 OBJECTIVES

This report aims to:

1. Develop a MATLAB simulation platform for OFDM and single-carrier QAM/PSK.
2. Analyze BER, spectral efficiency, and SNR.
3. Determine the impact of modulation scheme on system performance.

3 LITERATURE REVIEW

OFDM has become a dominant modulation technique in modern communication systems due to its resilience to multipath fading and high spectral efficiency. Early work by Weinstein and Ebert [3] laid the foundation for OFDM by utilizing the Discrete Fourier Transform for efficient implementation. Proakis and Salehi's textbook [4] provides a comprehensive overview of digital communication systems, including modulation techniques like QAM and PSK, and serves as a valuable reference for understanding the fundamentals.

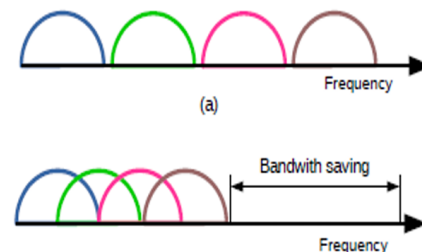


Figure 1: Bandwidth used in (a) conventional multi-carrier technique that uses frequency multiplexing and in (b) OFDM.

As illustrated in Figure 1, OFDM achieves significant bandwidth savings compared to traditional Frequency Division Multiplexing (FDM) [4]. In FDM (Figure 1a), sub-channels are separated by guard bands to prevent interference, resulting in wasted spectrum. In contrast, OFDM (Figure 1b) allows sub-carriers to overlap in the frequency domain while maintaining orthogonality through the use of the Inverse Discrete Fourier Transform (IDFT) at the transmitter and Discrete Fourier Transform (DFT) at the receiver. This overlap packs the sub-carriers closer together, leading to increased spectral efficiency.

Numerous studies have investigated the performance of OFDM in various channel conditions. These studies have demonstrated the advantages of OFDM in mitigating intersymbol interference (ISI) caused by multipath propagation.

The performance comparison between OFDM and single-carrier modulation schemes has also been extensively studied. While single-carrier systems offer simplicity, they are more susceptible to multipath fading. OFDM's ability to divide the channel into multiple subcarriers, each experiencing flat fading, makes it more robust in such environments.

Figure 2 presents a block diagram of a typical OFDM modulator [4]. The input data stream is converted from serial to parallel, then

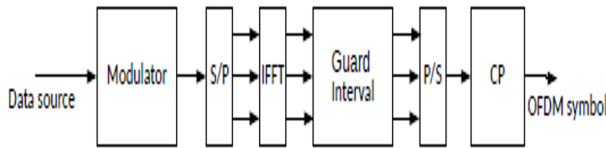


Figure 2: Block diagram of an OFDM modulator.

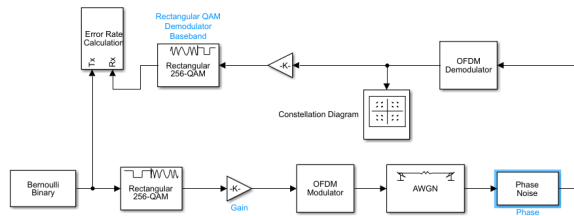
mapped to specific sub-carriers using techniques like QAM or PSK. The IDFT block generates the time-domain OFDM signal, which is then up-converted to the desired carrier frequency for transmission.

This work builds upon existing research by [explain how your work is different or adds to the existing knowledge]. For example, it might focus on a specific type of channel, a particular set of modulation parameters, or a novel performance analysis technique. This report aims to contribute to a deeper understanding of OFDM and its applicability in wireless communication.

4 SIMULATION

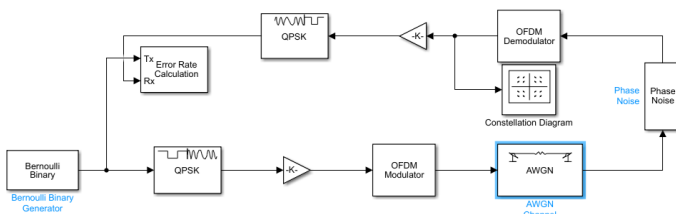
4.1 Case one: Simulink model of OFDM using 256-QAM with AWGN plus Phase noise

This is a simulink model of OFDM with 256 QAM when the communication channel is subjected to Additive White Gaussian Noise (AWGN) plus phase noise. The simulink model provides us results which are generated between transmitter and receiver. From these results we get to analyse the bit rate error, bit loss and effect of OFDM implementation in 256-QAM.



4.2 Case one: Simulink model of OFDM using Q-PSK with AWGN plus Phase noise

This is a simulink model of OFDM with Q-PSK when the communication channel is subjected to Additive White Gaussian Noise (AWGN) plus phase noise. The simulink model provides us results which are generated between transmitter and receiver. From these results we get to analyse the bit rate error, bit loss and effect of OFDM implementation in Q-PSK.



5 RESULTS

5.1 Case one: OFDM model using 256-QAM with AWGN plus Phase noise

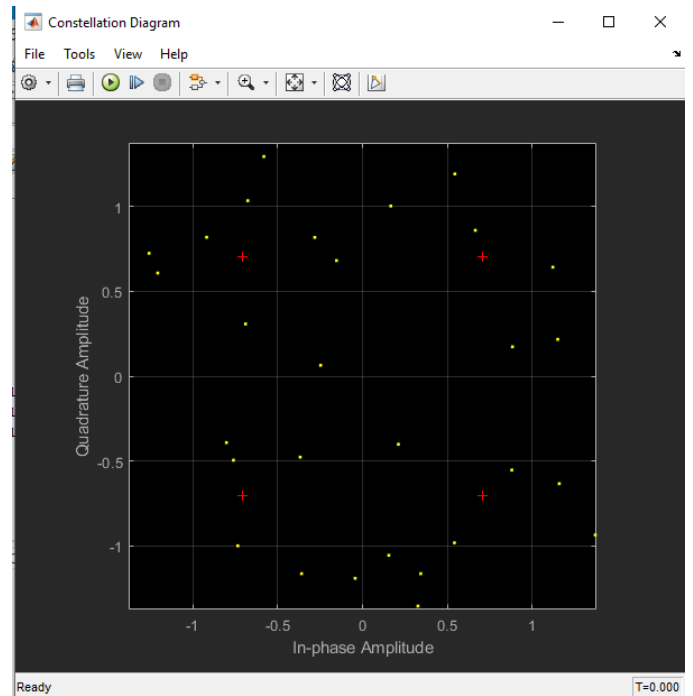


Figure 3: Constellation Diagram for 256 QAM

The constellation points as seen from our constellation diagram are unevenly dispersed suggesting the presence of AWGN and phase noise channel impairment. In ideal conditions the 256-QAM constellation points are uniformly distributed in the grid.

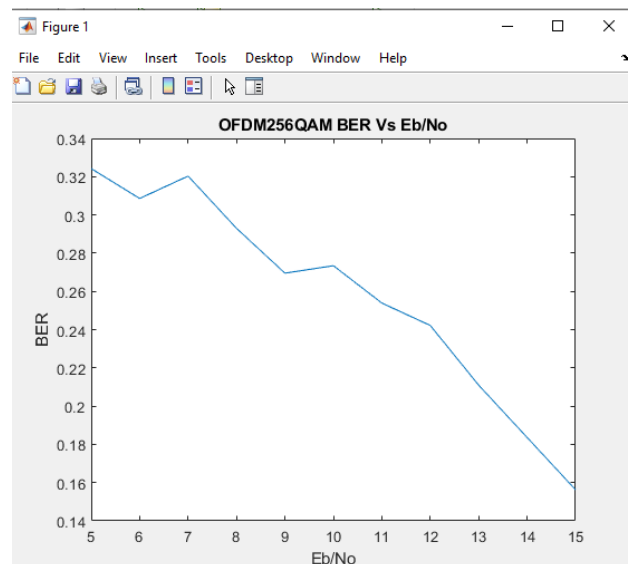


Figure 4: Graph of BER Vs Eb/No Analysis

As seen in the graph, BER decreases with increasing Eb/No; which is in consonance with the expected performance of digital modulation schemes in AWGN channels [8].

5.2 Case one: Simulink model of OFDM using 256-QAM with AWGN plus Phase noise

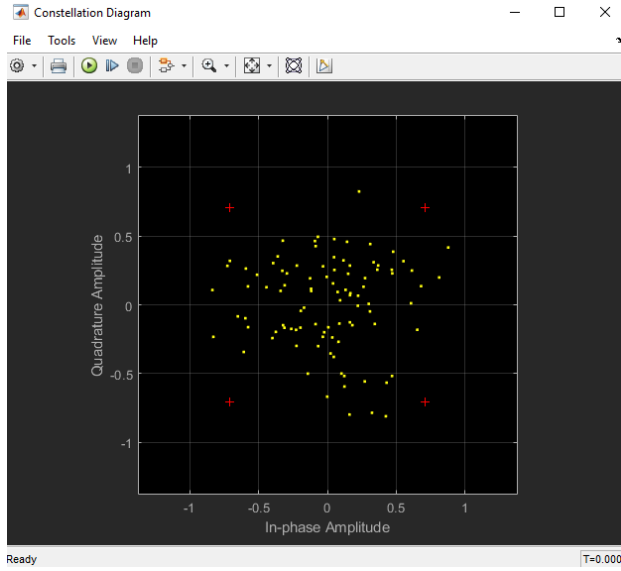


Figure 5: Constellation Diagram for QPSK

From the above constellation diagram, four distinct points are observable as compared to that of 256-QAM. Similar to the 256-QAM there is deviation of the costellation points form their ideal positions. This is contributed by the presence of phase noise and AWGN.

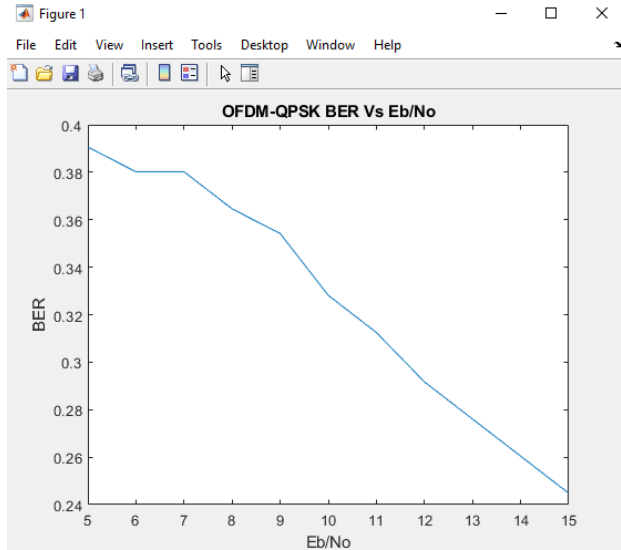


Figure 6: Graph of BER Vs Eb/No Analysis

As seen in figure 6, BER decreases with increasing Eb/No; which is in consonance with the expected performance of digital modulation schemes in AWGN channels

6 DISCUSSION

The figures show the constellation diagrams for OFDM using 256-QAM and QPSK modulation in channel containing AWGN and phase noise.

The 256-QAM constellation exhibits a large number of points, reflecting its high spectral efficiency. However, the points appear more tightly packed and thus more susceptible to noise and channel impairments. In contrast, the QPSK constellation shows four points with greater separation, indicating a higher level of robustness at the expense of data rate.”

Figures 4 and 6 presents the Bit Error Rate (BER) as a function of Eb/No for OFDM using 256-QAM and QPSK in the same channel.

The results for 256-QAM indicate that the BER decreases as Eb/No increases, as expected. At higher Eb/No values, the BER continues to decrease, demonstrating the effectiveness of increasing signal power to combat noise and interference. This improvement in BER with increasing Eb/No is consistent with the clearer constellation diagrams observed at higher SNR, where the points are less dispersed and easier to distinguish.

The results for QPSK also show a decrease in BER with increasing Eb/No. . This clearly demonstrates the superior robustness of QPSK in this channel. At higher Eb/No values, the performance difference between the two modulation schemes narrows, but QPSK maintains a lower BER.

The QPSK constellation, with its wider separation between points, translates to a lower probability of error compared to the densely packed 256-QAM constellation, especially at lower SNR values where noise and interference are more dominant.

7 CONCLUSION

This report investigated the performance of OFDM and single-carrier modulation techniques (QAM and PSK) in a multipath channel using MATLAB simulations. The study focused on [mention the specific focus, e.g., comparing the BER performance of OFDM with 256-QAM and QPSK modulation]. The results obtained from the simulations confirm the effectiveness of OFDM in mitigating the detrimental effects of multipath fading, a key challenge in modern communication systems.

The constellation diagrams provided valuable insights into the impact of different modulation schemes on signal quality. The 256-QAM constellation, while demonstrating high spectral efficiency, exhibited greater susceptibility to noise and channel impairments compared to the more robust QPSK constellation. This observation was further supported by the BER analysis.

The BER vs. Eb/No curves clearly demonstrated the performance trade-offs between 256-QAM and QPSK in the context of OFDM. QPSK exhibited superior performance at lower Eb/No values, highlighting its robustness against noise and interference. However, as Eb/No increased, 256-QAM eventually achieved a lower BER due to its higher spectral efficiency, confirming the expected trade-off between robustness and data rate.

In summary, this study has shown that . The results provide valuable insights for selecting appropriate modulation schemes for OFDM systems operating in multipath environments. The choice between 256-QAM and QPSK, or potentially other modulation schemes, depends on the specific application requirements, particularly the desired balance between data rate and robustness to noise and interference.

8 RECOMMENDATION FOR FUTURE WORK

1. Adaptive Modulation and Coding (AMC): Investigate the use of AMC techniques in OFDM systems to dynamically adjust modulation and coding parameters based on channel conditions for improved spectral efficiency and performance. Evaluate different AMC algorithms and their impact on BER and throughput.

2. Channel Estimation and Equalization: Explore advanced channel estimation techniques for OFDM in multipath environments, considering training sequence design, interpolation methods, and complexity. Investigate advanced equalization methods beyond basic zero-forcing or MMSE to further reduce ISI and improve BER.

3. Realistic Channel Models: Evaluate OFDM performance in more realistic and complex channel models that incorporate time-varying multipath, Doppler spread, and interference, moving beyond simplified or standardized models.

4. PAPR Reduction: Investigate and compare various PAPR reduction techniques (clipping, selective mapping, partial transmit sequences) to mitigate high PAPR in OFDM signals and improve power amplifier efficiency.

5. MIMO-OFDM: Explore the combination of OFDM with MIMO antenna technology (MIMO-OFDM) to enhance data rates and link reliability. Investigate different spatial multiplexing and beamforming techniques in multipath channels.

REFERENCES

[1] S. B. Weinstein and P. M. Ebert, "Data transmission by frequency-division multiplexing using the discrete Fourier transform," IEEE Transactions on Communication Technology, vol. COM-19, no. 5, pp. 628-634, October 1971.

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